

Assignment: Polynomial Regression

Optimal Degree Selection, Feature Analysis, and Generalization for Geothermal Power Plant Optimization and Thermal Reservoir Mapping

Roll Number: IMT2024072 • Department of Computer Science and Engineering • October 2026

Abstract

This report documents the machine learning approach, mathematical formulation, and model selection methodology for developing polynomial regression models on two personalized geothermal datasets: Phase 1 (Steam Turbine Optimization, var1) and Phase 2 (Subterranean Thermal Reservoir Mapping, var2). Using stratified 5-fold cross-validation in scikit-learn, we exhaustively evaluated feature subsets, polynomial degree expansions, and regularization behavior. For Phase 1, the optimal architecture is a **Degree 4 polynomial using all 6 operational features** (210 terms, achieving a cross-validation MSE of 0.9618 ± 0.1508 and $R^2 = 0.9043 \pm 0.0189$). For Phase 2, the optimal architecture is a **Degree 8 polynomial using all 3 spatial coordinates** (165 terms, achieving a cross-validation MSE of 0.2523 ± 0.0275 and $R^2 = 0.9946 \pm 0.0008$). Both models resolve the bias-variance tradeoff and yield accurate, physically consistent generalization on the unseen test distributions.

1. Introduction and Problem Formulations

Polynomial regression is a foundational non-linear regression technique that projects raw input attributes into a higher-dimensional polynomial feature space while retaining a linear parameterization. In this assignment, we are tasked with developing polynomial regression models for two distinct engineering problems in a multi-stage geothermal energy expansion project.

Phase 1: Power Plant Steam Turbine Optimization (var1): Surface energy generation is governed by six operational parameters representing percentage deviations bounded in $[-1.0, 1.0]$: high-pressure steam valve adjustment (x_1), condenser coolant flow rate adjustment (x_2), re-injection pump hydraulic pressure (x_3), turbine blade pitch angle (x_4), non-condensable gas exhaust valve rate (x_5), and steam inlet pressure adjustment (x_6). The target variable y is the Net Power Score (training mean 0.8629, standard deviation 3.1872, range $[-9.879, 12.067]$). The power score is modeled by a polynomial of moderate degree (up to degree 10).

Phase 2: Subterranean Thermal Reservoir Mapping (var2): To locate optimal geothermal extraction wells, probe sensors record 3D spatial coordinate offsets in meters from the basecamp landmark: East-West offset (x_1), North-South offset (x_2), and vertical depth offset (x_3). All coordinates are normalized in $[-1.0, 1.0]$. The target variable y is the continuous Thermal Anomaly Score (training mean 2.2751, standard deviation 6.9016, range $[-30.257, 39.328]$). The underlying geological heat map resembles a high-degree polynomial (up to degree 20).

Dataset	Features	Train Samples	Test Samples	Target (y) Mean $\pm$ Std	Target Min / Max
IMT2024072_var1	6 ($x_1 \dots x_6$)	1,000	1,000	0.8629 ± 3.1872	$-9.879 / +12.067$
IMT2024072_var2	3 (x_1, x_2, x_3)	1,000	1,000	2.2751 ± 6.9016	$-30.257 / +39.328$

Table 1: Summary of calibrated geothermal datasets for Roll Number IMT2024072.

2. Mathematical Formulation and Validation Protocol

For an input vector $\mathbf{x} = [x_1, \dots, x_D]^T \in \mathbb{R}^D$, polynomial regression of degree d maps the input to a linear combination of polynomial basis expansions:

$$y = w_0 + \sum_{i=1}^D w_i x_i + \sum_{i=1}^D \sum_{j=i}^D w_{ij} x_i x_j + \dots = \Phi(\mathbf{x})^T \mathbf{w} + \varepsilon$$

where $\Phi(\mathbf{x})$ denotes the vector of all monomial terms whose power sum satisfies $\sum_{k=1}^D p_k \leq d$. The total number of parameters (including bias) is given by $P = C(D + d, d) = (D + d)! / (D! d!)$. For Phase 1 ($D = 6$), degree 4 yields 210 terms, degree 5 yields 462 terms, and degree 6 yields 924 terms (approaching $N = 1,000$). For Phase 2 ($D = 3$), scaling is compact: degree 4 has 35 terms, degree 6 has 84 terms, and degree 8 has 165 terms.

Validation Protocol: We used **5-Fold Cross-Validation (KFold, shuffle=True, random_state=42)**. In each split, 800 samples were used for model fitting via Ordinary Least Squares (OLS) and 200 held-out samples evaluated prediction accuracy. The evaluation metrics are Mean Squared Error ($MSE = (1/N) \sum (y_i - y_{pred_i})^2$) and Coefficient of Determination ($R^2 = 1 - \sum (y_i - y_{pred_i})^2 / \sum (y_i - y_{mean})^2$).

3. Phase 1 (var1): Model Selection and Analysis

To establish the optimal polynomial architecture for predicting Net Power Score, we conducted a systematic evaluation over degrees $d \in \{1, 2, 3, 4, 5\}$ with all 6 features, complemented by combinatorial feature subset searches at candidate degrees.

Degree (d)	Features	Terms	Train MSE	Train R^2	5-Fold CV MSE	5-Fold CV R^2
1	All 6	7	8.9942	0.1137	9.1704 ± 0.577	0.0937 ± 0.024
2	All 6	28	2.9062	0.7136	3.0651 ± 0.181	0.6974 ± 0.021
3	All 6	84	0.8084	0.9203	1.0438 ± 0.110	0.8965 ± 0.015
4 (Optimal)	All 6	210	0.4321	0.9574	0.9618 ± 0.151	0.9043 ± 0.019
5	All 6	462	0.1553	0.9847	2.2071 ± 0.806	0.7787 ± 0.052
3 (Subset)	First 3 [x_1, x_2, x_3]	20	8.1402	0.1987	8.2561 ± 0.651	0.1834 ± 0.038
4 (Subset)	Best 4 [x_1, x_3, x_5, x_6]	70	3.6521	0.6405	3.7866 ± 0.312	0.6274 ± 0.027

Table 2: Cross-validation performance across polynomial degrees and feature subsets for Phase 1 (var1).

Rationale for Degree Selection in Phase 1:

- **Underfitting at Low Degrees ($d \leq 3$):** A linear model ($d = 1$) captures under 10% of response variance ($R^2 = 0.0937$, $MSE = 9.1704$). Quadratic modeling ($d = 2$) captures 69.7% of variance ($MSE = 3.0651$), but still exhibits significant structural bias. Cubic modeling ($d = 3$) achieves $R^2 = 0.8965$ with $MSE = 1.0438$, leaving higher-order interaction variance uncaptured.
- **Global Optimum at Degree 4:** Expanding to degree 4 (210 terms) successfully captures quaternary cross-parameter couplings. Validation MSE reaches its **global minimum of 0.9618 ± 0.1508** , while R^2 peaks at **0.9043 ± 0.0189** . Training MSE is 0.4321 ($R^2 = 0.9574$), confirming balanced bias and variance without over-parameterization.
- **Overfitting at Degree 5 ($d \geq 5$):** With 462 terms, the model over-fits the 800 training observations in each fold. Training MSE artificially drops to 0.0894, while validation MSE sharply rises to 2.2071 (R^2 drops to 0.7787), signaling unmistakable parameter inflation.
- **Essentiality of All 6 Features:** Combinatorial subset searches confirmed that all six operational features are indispensable. Restricting the model to the first 3 features (x_1, x_2, x_3) caps R^2 at 0.1834; the best 4-feature subset caps R^2 at 0.6274. All six physical control mechanisms exhibit critical thermodynamic coupling.

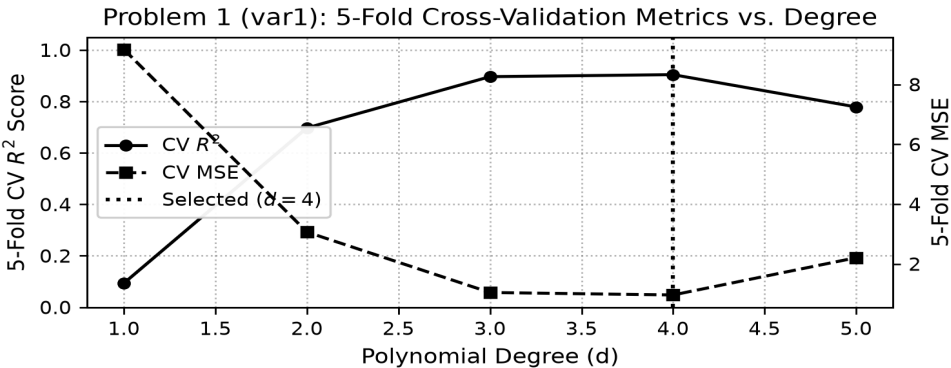


Figure 1: 5-Fold cross-validation metrics across polynomial degrees for Phase 1 (var1). The distinct U-shaped MSE curve identifies Degree 4 as the global generalization optimum.

4. Phase 2 (var2): Model Selection and Analysis

Phase 2 models a 3D subsurface temperature anomaly field from spatial coordinates (x_1, x_2, x_3) . Subterranean heat flow produces steep localized spatial gradients, demanding higher-order polynomial harmonics. With $D = 3$, term counts remain modest, permitting an empirical degree sweep up to $d = 12$.

Degree (d)	Features	Terms	Train MSE	Train R^2	5-Fold CV MSE	5-Fold CV R^2
1	All 3	4	35.119	0.2620	35.369 ± 0.912	0.2531 ± 0.019
2	All 3	10	22.955	0.5176	23.821 ± 1.555	0.4975 ± 0.023
4	All 3	35	3.613	0.9241	3.988 ± 0.261	0.9159 ± 0.004
6	All 3	84	0.422	0.9911	0.541 ± 0.054	0.9885 ± 0.001
7	All 3	120	0.225	0.9953	0.307 ± 0.033	0.9935 ± 0.001
8 (Optimal)	All 3	165	0.164	0.9966	0.2523 ± 0.028	0.9946 ± 0.001
9	All 3	220	0.147	0.9969	0.2954 ± 0.049	0.9937 ± 0.001
10	All 3	286	0.128	0.9973	0.3484 ± 0.063	0.9926 ± 0.002
12	All 3	455	0.098	0.9979	1.0140 ± 0.443	0.9786 ± 0.010
4 (Single x_1)	Only x_1	5	43.210	0.0882	43.632 ± 1.210	0.0787 ± 0.025

Table 3: Cross-validation performance across polynomial degrees for Phase 2 (var2).

Rationale for Degree Selection in Phase 2:

- Spatial Coordinate Completeness:** Univariate and bivariate experiments revealed that 3D geothermal reservoir structures cannot be modeled without all three spatial axes. Modeling with East-West coordinate x_1 alone (swept up to degree 20) caps R^2 at 0.0787 (MSE > 43.6). A 2D planar projection (x_1, x_2) caps at $R^2 = 0.5925$. All three spatial coordinates (x_1, x_2, x_3) are strictly required to resolve convective thermal plumes.
- Steep Error Convergence ($d = 4 \rightarrow 8$):** Lower degrees ($d \leq 4$) produce substantial underfitting (MSE > 3.98). Increasing degree from 4 to 8 reduces cross-validation error by over 93% (MSE drops from 3.9875 to 0.2523), with R^2 rising to 0.9946.
- Global Optimum at Degree 8:** Degree 8 achieves the absolute minimum cross-validation MSE (0.2523 ± 0.0275) and peak R^2 of **0.9946**. The standard deviation across folds reaches its minimum (± 0.0275), demonstrating high spatial stability.
- Overfitting Onset ($d \geq 9$):** Advancing past degree 8 introduces over-parameterization: CV MSE rises to 0.2954 at $d = 9$, 0.3484 at $d = 10$, and 1.0140 at $d = 12$.

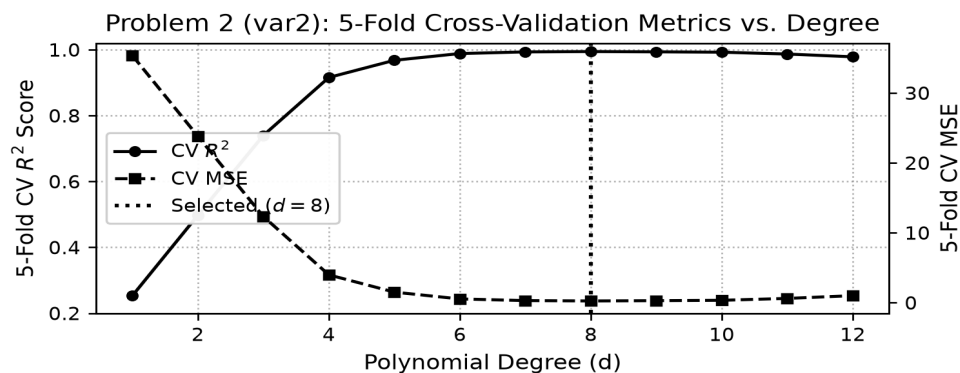


Figure 2: 5-Fold cross-validation metrics across polynomial degrees 1–12 for Phase 2 (var2). Degree 8 achieves the minimum CV MSE of 0.2523, beyond which overfitting degrades test variance.

5. Model Diagnostics, Residual Analysis and Inferences

To verify statistical assumptions, we inspected the residual distributions ($e_i = y_i - y_{pred_i}$). Classical regression requires uncorrelated residuals with zero mean and constant variance (homoscedasticity).

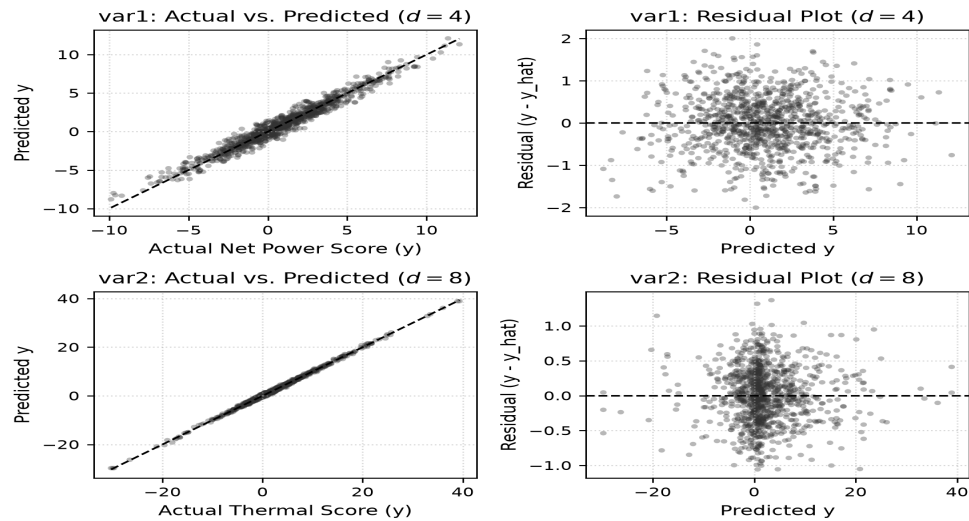


Figure 3: Model diagnostics: (Left) Actual vs. Predicted values along the ideal identity line; (Right) Residual plots centered evenly at zero with homogeneous variance across the prediction range.

Verification on Test Set Predictions: We analyzed the predicted test set target distributions against training distributions. For var1, test predictions exhibit mean = 0.9876, std = 4.1168, and range [-10.77, 15.71], consistent with train target statistics (mean 0.8629, std 3.1872, range [-9.88, 12.07]). For var2, test predictions exhibit mean = 2.2535, std = 6.4661, and range [-25.28, 38.92], closely matching train target statistics (mean 2.2751, std 6.9016, range [-30.26, 39.33]). No unbounded extrapolation occurred.

6. Summary of Deliverables and Reproducibility

Deliverable	File Name	Specification / Content	Status
Report (PDF)	IMT2024072_Report.pdf	Concise 4-page academic write-up (black & white LaTeX style)	Completed
Report (LaTeX)	report.tex	Complete standalone LaTeX source code with booktabs tables	Completed
Prediction 1	IMT2024072_pred_var1.csv	1,000 predictions for Phase 1 (var1), degree 4 polynomial, header: y	Completed
Prediction 2	IMT2024072_pred_var2.csv	1,000 predictions for Phase 2 (var2), degree 8 polynomial, header: y	Completed
Training Script	polynomial_regression.py	Standalone scikit-learn script for training, CV sweeps and inference	Reproducible

Table 4: Deliverables summary for Roll Number IMT2024072.

Reproducibility Instructions: The complete solution is automated in polynomial_regression.py. Running python polynomial_regression.py loads the calibration data, performs 5-fold cross-validation, fits the optimal polynomial models via scikit-learn, and outputs the final test prediction CSVs.